# 04-07-2025

Final: 5-1-2025
12-3 PM
A400 (?)

$y, a \Rightarrow$

| | $a_1$ | $a_2$ | $a_3$ |
|---|---|---|---|
| $y_L$ | $\Gamma_{11}$ | $\Gamma_{12}$ | $\Gamma_{13}$ |
| $y_H$ | $\Gamma_{21}$ | $\Gamma_{22}$ | $\Gamma_{23}$ |

$\sum_i \sum_j \Gamma_{ij} = 1$

Measure is static, position can change

↳ stationary due to measure not changing

Hansen and Imromorogulu (1992)

- $\varepsilon = \{e, u\}$
- $\pi(\varepsilon' | \varepsilon) = \begin{pmatrix} P_{uu} & P_{ue} \\ P_{eu} & P_{ee} \end{pmatrix}$
- $u(c_t) = \frac{c_t^{1-\sigma}}{1-\sigma}$

e: w (labor = 1)
u: b (unemp benefit lump sum)

$$V(a, \varepsilon) = \max_{c, a'} \{u(c) + \beta \sum_{\varepsilon'} \pi(\varepsilon' | \varepsilon) V(a', \varepsilon')\}$$

s.t.

$$c = y + a(1+r) - a'$$

$$y = \begin{cases} w(1-\tau) & \text{if } \varepsilon = e \\ b & \text{if } \varepsilon = u \end{cases}$$

Given prices $\{w, r\}$ and policy vars $\{\tau, b\}$,
consumer solves

define:

$$c = \Psi(a,\varepsilon) \;,\; a' = g(a,\varepsilon) \Rightarrow v(a,\varepsilon)$$

Production Sector:

$$Y = K^{\alpha}N^{1-\alpha}$$

$$\Pi = K^{\alpha}N^{1-\alpha} - wN - rK - \delta K \longrightarrow \max_{K,N}$$

$$\Rightarrow r+\delta = \alpha K^{\alpha-1}N^{1-\alpha} = \alpha\left(\frac{K}{N}\right)^{1-\alpha}$$

$$\Rightarrow w = (1-\alpha)K^{\alpha}N^{-\alpha} = (1-\alpha)\left(\frac{K}{N}\right)^{\alpha}$$

Government:

Total Benefits $:= B = T :=$ Tax Revenue

Stationary Distribution:

$F(e,a)$ ; $F(u,a) \rightarrow$ CDF
↳ also $F(\varepsilon, a)$ works
↳ asset distribution of employed, unemployed

$f(e,a)$ ; $f(u,a) \longrightarrow$ PDF
↳ $f(\varepsilon,a)$
→ number unemployed should be same with potential for movement within

Going to treat "b" as exog.

Stationary C.E. (SCE)

- <u>given b</u>, SCE for this economy is

- policy functions AND value functions
  - $\psi(a,\varepsilon)$, $g(a,\varepsilon)$, $v(a,\varepsilon)$
- set of prices $\{w,r\}$
- gov't policy variable $\tau$
- aggregate variables $\{K, N, C, T, B\}$
- distribution of agents over state space $F(a,\varepsilon)$

s.t.

1. policy f(·): $\psi(\cdot)$, $g(\cdot)$, and $v(\cdot)$ solve consumer problem <u>given $\{w,r\}$, $\{\tau, b\}$</u>

2. given $\{w,r\}$, firm optimizes

3. $\{T, B\}$ satisfy MCC's

   - Labor: $N = \int_a f(a,e)\,da$

   - Capital: $K = \int g(a,e) f(a,e)\,da + \int g(a,u) f(a,u)\,da$

   - Consumption Good: $C = \int \psi(a,e) f(a,e)\,da + \int \psi(a,u) f(a,u)\,da$

     $= Y - I$

     $I = K' - (1-\delta)K = \delta K$

4. given $\{b, \tau\}$, the gov't budget is balanced s.t. $B = T$, $B = (1-N)b$; $T = \tau N w$ $\therefore (1-N)b = \tau N w$

5. distribution of agents over state space is stationary: $F(a', \varepsilon') = \sum_{\varepsilon} \pi(\varepsilon' | \varepsilon) F(g^{-1}(a', \varepsilon), \varepsilon)$
   - akin to law of motion of distribution

$g^{-1}(\cdot) :=$ $e, a_m \longrightarrow a_m$; $u, a_n \longrightarrow a_m$

Calibration:

1 period = 6 weeks

$\alpha = 0.36$

risk aversion, $\sigma = 2$

$\beta = 0.995$ (match int. rate of 0.04)

$$\begin{pmatrix} P_{uu} & P_{ue} \\ P_{eu} & P_{ee} \end{pmatrix} = \begin{pmatrix} 0.5 & 0.5 \\ 0.0435 & 0.9565 \end{pmatrix}$$

Average duration of unemp: 2 periods

Average unemp. rate = 0.08

$b = 0.25\, w^{Rep}$

To find Rep Agent:

- $N = 1$ (only employed)
- $\beta(1+r^{Rep}) = 1$ (flat consumption)

$r^{Rep} = \frac{1}{\beta} - 1$

$r^{Rep} + \delta = \alpha (K)^{\alpha-1} \Rightarrow K^{rep} = \left[(r^{Rep} + \delta) \cdot \alpha^{-1}\right]^{\frac{1}{1-\alpha}}$

$w^{Rep} = (1-\alpha)(K^{rep})^{\alpha}$

Computational Roadmap:

- b_exog

1: guess $r \Rightarrow$ find $\frac{K}{N} = \left[\frac{r+\delta}{\alpha}\right]^{\frac{1}{1-\alpha}} \Rightarrow$ find $w$

2: $\tau$ - gov't constraint: $\tau w N = b(1-N)$

3: N is decided by transition matrix (exog)
↳ N represents stationary distribution $\begin{pmatrix} N \\ 1-N \end{pmatrix}$